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# Estimating on-farm genotypic performance and variability using ranking data

[version 1]

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## Abstract

Plant breeding research has chiefly relied on on-station experiments for evaluating varietal performance. Nevertheless, these have clear limitations in representing on-farm growing conditions and farmers' preferences, potentially leading to the misspecification of breeding targets. Recent work has demonstrated the potential of using structured on-farm verification trials combined with ranking data to support farmers evaluate varieties in their fields, providing representative information on their needs. However, there has been a lack of scalable methods to quantify differences in genetic performance and assess the strength of the genetic signal in such trials. In this paper, we present a two-step procedure for analyzing trials based on ranking data, allowing the estimation of genetic parameters. This procedure follows a common approach in quantitative genetics, where parameters are estimated using tables of estimated genotypic means and their variances. In our case, these

tables are derived from Thurstonian and/or Plackett-Luce models, which assume that rankings reflect an underlying continuous trait linked to genotypic performance. Using simulated data, we validated that genotypic mean estimates from ranking analyses are linearly related to those from quantitative trait analyses, and that their variances adequately capture associated uncertainty. We further confirm that a second-step analysis of ranking data using weighted mixed-effect models yields accurate estimates of variance components. Analyses of observed data on groundnut, maize, and sweetpotato demonstrate the applicability of this approach and confirm that ranking data provide reliable estimates of genetic parameters. We argue that this framework can be scaled to obtain genotypic performance estimates from multi-trial on-farm data.

**Keywords**
Ranking data, on-farm trials, plant breeding, Plackett-Luce model, Thurstonian model, and tricot approach

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## Key message

Our scalable two-step method estimates genotypic performance and genetic parameters from ranking data, producing reliable results comparable to quantitative analyses, enabling integration of this type of data into breeding pipelines.

## 1. Introduction

Plant breeders need to assess newly developed genotypes, which is typically done using on-station trials, involving intensive phenotyping protocols and covering a limited set of environments. On-farm testing serves as a complementary approach, allowing genotype performance and farmer preferences to be assessed under actual growing conditions (Werner et al., 2024; Worku et al., 2020). On-farm testing is widely used, but is especially critical for breeding programs that target diverse, and low-input environments. Recently, a scalable citizen-science-based approach has been developed as a cost-effective method for on-farm testing of genotypes (van Etten, Beza, et al., 2019a). In this approach, known as the triadic comparisons of technologies (tricot), each farm is treated as an incomplete block, where the performance of three genotypes is ranked by the farmer from the best to the worst. Tricot is easy to implement for medium- and small-scale farmers and improves reliability in evaluating traits such as yield, color, taste, and marketability (de Sousa et al., 2024). However, a current limitation of tricot is the lack of methodologies to estimate the components of genetic and non-genetic variation from trials based on ranking evaluations, similar to those applied in quantitative genetic analyses of continuous traits. Consequently, this limitation restricts the potential use of on-farm trials based on ranking data for assessing genetic progress, evaluating trial reliability, and supporting decision-making regarding variety release.

It is possible to obtain quantitative estimates of genotype performance from ranking data by treating rankings as expressions of an underlying continuous trait. Two commonly used models, Thurstone Case V ('Thurstonian model' hereafter) and Plackett-Luce, adopt this approach by assuming respectively a normal and a Gumbel distribution (type I extreme value), for the underlying trait (Luce, 1959; Thurstone, 1927; Yu et al., 2019). The Plackett-Luce model has been employed for evaluating and summarizing relative genotypic performance based on farmer-generated rankings (de Sousa et al., 2021; van Etten, de Sousa, et al., 2019b), although it has not been used to estimate genetic parameters of interest. In contrast, the Thurstonian model is a more natural choice based on classical quantitative genetics, where continuous traits are typically assumed to follow a normal distribution (Falconer, 1996). This model has been incorporated into Bayesian mixed-model frameworks for the quantitative genetic analysis of ranked traits (Gianola, 1982; Ricard & Legarra, 2010), primarily in the context of animal breeding (Varona & Legarra, 2020) with only a few applications in crop experimentation (Halekoh & Kristensen, 2008). Although the potential use of Thurstonian model in plant breeding applications, the current implementations are computationally demanding, difficult to adapt to specific data structures, and often restrictive to integrate into standard analytical pipelines of plant breeders. In current practice, plant breeders often use tables of estimated genotypic means and their variances as the main input for quantitative genetic analyses, which facilitates the aggregation and simultaneous analysis of multiple trials. However, a similar approach has not been implemented or tested in breeding experiments that use ranking data.

In this study, we propose and validate a simple and scalable two-step procedure for using ranking data as input for a quantitative genetic analysis of on-farm trial data. Currently, tricot trials data analysis is typically reported in tables of estimated genotypic means and their associated variances, derived from ranking data analyzed with the Plackett–Luce model. Similar tables are generated in quantitative traits' analysis and have been successfully used in downstream analyses to estimate genetic parameters from aggregated trials (networks) using a two-stage approach analysis (Smith et al., 2001; Verbyla, 2023). Our procedure follows the same principle, except that in the initial step, the tables of estimated genotypic means and variances are obtained from ranking data analysis and expressed on an underlying continuous scale. After obtaining these estimates, the procedure for estimating genetic parameters, such as heritability, genetic standard deviation, and genetic correlations, based on linear mixed models is similar to the analysis of quantitative traits.

We propose that our approach for analyzing experiments with ranking data can perform comparably to conventional quantitative-trait analysis, provided that the genotypic mean estimates derived from the ranking data and their associated variation are reliable. To test under which experimental conditions both criteria are met, we simulate and analyze tricot data for which the link between the underlying continuous trait and the ranking data is known. This analysis includes a comparison between the Thurstonian model, which meets the assumption of normality typically required for most quantitative trait analyses, and the Plackett–Luce model, which is the default for analyzing ranking data in tricot. Lastly, we evaluate our approach on observed data in which yield was measured and ranked on the same farms.

Our objectives are: (1) to define a robust and scalable two-step procedure for quantitative genetic analysis of ranking data; (2) to evaluate the performance of the estimates produced by this framework through a simulation study; and (3) to compare estimates based on ranking and quantitative data for the same trait using observed datasets. We thereby aim to

lay the methodological foundation for incorporating ranking data from on-farm variety trials into pipelines for the analysis of single and multiple trials. This will give plant breeders a tool for reliably estimating on-farm genotypic performance and variability, helping guide decisions that respond to farmers' needs.

## 2. Material and methods

### 2.1 Tricot trials

The motivation of our contribution is to enable the use of ranking data in plant breeding, specifically because of the prevalence of this type of data in the tricot approach. Tricot follows a citizen science approach for on-farm testing, where each farm is treated as an incomplete block. On each farm, three genotypes drawn from a larger pool are cultivated in separate plots and evaluated by the farmer, who ranks them from best to worst, based on observed performance. These evaluations are conducted independently for multiple traits, including overall performance. Limiting comparisons to three genotypes simplifies the evaluation process and reduces error, providing a practical way for small- and medium-scale farmers to give feedback to breeders and scientists (de Sousa et al., 2024). Tricot uses a forced-choice approach, meaning that no ties are allowed. Each block is observed and evaluated by the farmer who manages it, ensuring that the data reflects direct field experience. Repeated evaluations can be conducted, but each block corresponds to one evaluator.

### 2.2 Thurstonian model for ranking data analysis

For a quantitative genetic analysis based on ranking data, we focus on probabilistic models that enable continuous estimation of genotypic performance from observed rankings. Under the Thurstonian model, rankings are treated as outcomes of a latent process in which the observed ranks represent underlying continuous utilities. In a given occurrence, the ordering of these utilities is directly linked to the observed ranking (Thurstone, 1927; Yu et al., 2019). These utilities follow an associated probabilistic distribution whose parameters can be estimated from observed rankings, enabling subsequent quantitative genetic analyses.

Building on this framework, we adapt the Thurstonian approach to experiments in which genotypes are evaluated using ranking data, defining a continuous variable $y_{jk}$ that represents the utility of genotype $j$ ($j = 1, 2, \ldots, v$), as assessed by the evaluator/farmer/ $k$ ($k = 1, 2, \ldots, t$):

$$y_{jk} = \mu_j + \varepsilon_{jk} \tag{1}$$

Where $\mu_j$ is the mean utility of genotype $j$ across all evaluations, and $\varepsilon_{jk}$ is a random component capturing perception error and environmental variation. Under the Thurstonian model, $\varepsilon_{jk}$ is normally distributed with mean 0 and variance $\sigma^2_\varepsilon$. The main effect of the evaluator does not influence the observed rankings and is therefore is 0 for in Equation 1.

From this point forward, we refer to the mean utilities of genotypes $\mu_1, \mu_2, \ldots$, as genotypic utility means.

The probability $P$ of obtaining a particular ranking $R_k$ under the Thurstonian model is defined using an order statistic model (Yu et al., 2019). We illustrate this probability with a simple example based on a tricot experiment. Consider a set of genotypes $geno\,1, geno\,2, \ldots, geno\,v$. Suppose farmer $k$ evaluates an incomplete block $S_k$ containing three of these genotypes: $\{geno\,1, geno\,5, geno\,8\}$. Farmer $k$ provides the ranking $R_k : geno\,5 > geno\,8 > geno\,1$, meaning that $geno\,5$ was the best and $geno\,1$ the worst. Following (Böckenholt, 1992) this probability is defined as a function of the underlying latent utilities as follows:

$$P(R_k : geno\,5 > geno\,8 > geno\,1 \,|S_k = \text{geno}\,1, \text{geno}\,5, \text{geno}\,8) = P(y_{5k} > y_{8k} > y_{1k}) = P\{(y_{5k} - y_{8k} > 0) \cap (y_{8k} - y_{1k} > 0)\} \tag{2}$$

The joint distribution of $y_k = (y_{5k}, y_{8k}, y_{1k})'$ follows a multivariate normal distribution with mean vector $\mu_{y_k}$ and covariance matrix $\Sigma_{y_k}$. By making the change of variables $U_{1k} = y_{5k} - y_{8k}$, $U_{2k} = y_{2k} - y_{1k}$, the joint distribution $U_k = (U_{1k}, U_{2k})'$, follows a bivariate normal distribution with mean vector $\mu_{U_k}$ and covariance matrix $\Sigma_{U_k}$. This definition of $U_k$ is valid for any set of genotypes.

Thus, probability $P$ can be calculated as:

$$P(\mathrm{R_k}|S_k) = P\{(U_{1k} > 0) \cap (U_{2k} > 0)\} = \int_0^\infty \int \phi\left(U_k : \mu_{U_k}, \Sigma_{U_k}\right) dU_{1k} dU_{2k} \tag{3}$$

Where $\phi$ is the density function of a bivariate normal distribution. The specification of $\mu_{U_k}$ and $\Sigma_{U_k}$, for the example mentioned, the corresponding values are provided in subsection 1.1 of the Supplementary Information. For the general definition of $P$ for comparisons of more than three items refer to Böckenholt (1992) and Tavernier (1991).

*2.2.1 Estimation of utility-based genotypic means, variances, and quasi-variances*

The probability $P$ of a ranking occurrence, as defined above, is used to construct the log-likelihood function $\ell$ for estimating the genotypic utility means $\widehat{\mu}_1, \widehat{\mu}_2, \ldots,$. These estimations are obtained from the maximization of $\ell$.

$$\ell(\mu_1,\mu_2, \ldots) = c + \sum_{k=1}^{t} \log\left(P(R_k|S_k)\right) \tag{4}$$

Where $c$ is a constant, $R_k$ is the ranking provided by the farmer $k$, and $S_k$ is the subset of genotypes evaluated by farmer $k$. To ensure identifiability, two restrictions are imposed. First, $\mu_1 = 0$, which is typically done applied using the genotypic mean of a check variety. This restriction provides a relative scale for all genotypic utility mean estimates (Böckenholt, 1992; Thurstone, 1927; Turner et al., 2020). Second, the variance of the random component of equation 3 is fixed to $\sigma_\varepsilon^2 = 1$. Fixing this variance to 1 expresses utilities in units of residual standard deviation (Gianola & Simianer, 2006). This interpretation links the underlying utility scale to the quantitative metric scale of the same trait. For example, between the analysis of yield ranked and yield measured in tons per hectare (tons/ha), the utilities assumed by the Thurstonian model are given in units of residual standard deviation of yield in tons per ha.

As we will discuss in the next section, both the estimates of genotypic utility means and their associated variances are crucial for estimating genetic parameters with ranking data. In the Thurstonian model, obtaining the covariance matrix of the genotypic utility mean estimates, $\widehat{\Sigma}_{(\widehat{\mu}_1,\widehat{\mu}_2,\ldots)}$ is not straightforward, due to the complexity of partial derivatives of $\ell$. To address this, we use an asymptotic approximation based on the observed Fisher information matrix (Efron & Hinkley, 1978), obtained from the Hessian matrix, which is derived within the maximization $\ell$.

The variance for the genotypic utility mean estimates are obtained directly from the diagonal elements of $\widehat{\Sigma}_{(\widehat{\mu}_1,\widehat{\mu}_2,\ldots)}$. However, due to the constraint imposed on the reference genotype $\mu_1 = 0$, only *v-1* variances are estimated, and these estimates depend on the choice of reference genotype. In other words, if we change our genotype reference, the estimates for all variances will change. Quasi-variances (qV) provide an alternative approach for expressing the uncertainty of genotypic utility mean estimates. Quasi-variance estimates are independent of a reference genotype and facilitate inference analysis on the performance of different genotypes (Firth, 2003; Ridout, 1989; Turner et al., 2020). The estimated qV $q\widehat{V(\widehat{\mu}_1)}, q\widehat{V(\widehat{\mu}_2)}, \ldots$ are derived from the full estimated covariance matrix $\widehat{\Sigma}_{(\widehat{\mu}_1,\widehat{\mu}_2,\ldots)}$, which is feasible for analyses using the Thurstonian model.

## 2.3 Comparison with the Plackett-Luce models

The Plackett–Luce model has so far been the default approach for analyzing tricot data and follows the same principle as the Thurstonian model in defining underlying utilities associated with the ranked items (Brown et al., 2022; de Sousa et al., 2020). Similarly, these utilities can be expressed as a function of a mean utility and a random error, equivalent to Equation (1). The advantages of the Plackett-Luce model are that it facilitates computation, that it obeys the Luce Axiom (independence of irrelevant alternatives), and that it is available in many tools for data analysis, including an R package (Turner et al., 2020).

The main difference of relevance here is that the Plackett–Luce model assumes that $y_{jk}$ follows a Gumbel distribution (type I extreme value), while the Thurstonian model assumes a normal distribution (Allison & Christakis, 1994; Turner et al., 2020; Yellott, 1977). This affects the scale of the estimates derived from the model. If the underlying trait is normally distributed, as is typically assumed in quantitative genetics, estimates from the Plackett–Luce model are biased. This bias could be corrected by a scaling factor but this depends on the genetic-to-residual standard deviation ratio and on the number of replicates (Supplementary 1.3 and Fig. S1). In contrast, for traits with an underlying normal distribution, estimates from the Thurstonian model are unbiased and do not depend on any additional parameters. Comparisons of the estimates derived from these two models for ranking data analysis are presented in the Results section, and a scaling factor to correct Plackett–Luce estimates when the underlying trait is normally distributed is discussed in Supplementary 1.3.

## 2.4 Our framework for quantitative genetic analysis of ranking data

We adapt a two-step approach commonly used in quantitative genetic analyses (Piepho et al., 2012; Verbyla, 2023) to a framework that handles ranking data. In the first step, genotypic means and quasi-variances (qV) are estimated on the utility scale. These estimates are then used in a weighted linear mixed-effects model to estimate genetic parameters in the second step (Figure 1). We illustrate our framework using a single-environment trial, but it can be readily extended to

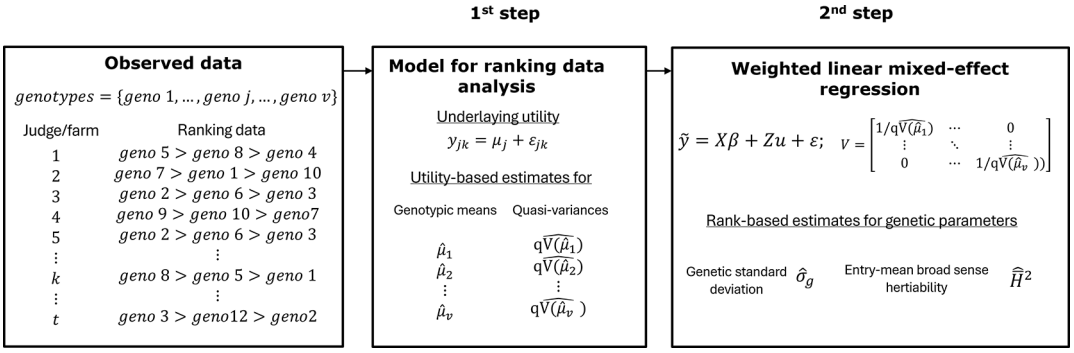



**Figure 1. Two-step procedure for estimating genotypic means, genetic standard deviation, and heritability using ranking data.** Here, $\widetilde{y}$ is a vector of genotypic utility mean estimates; $\beta$ and $u$ are vectors of fixed and random effects, with design matrices, $X$ and $Z$ respectively; and $\varepsilon$ is the residual vector. $V$ is matrix of weights used to fit the mixed-effect model.

multi-environment analyses, provided that tables of genotypic means and their corresponding qV estimates are available for each environment.

In the first step of our framework, we move from a discrete ordinal space to a continuous one, using the ranking data as an input that allows to quantify the performance of the genotypes evaluated. Here, we use the Thurstonian model to derive the genotypic utility mean estimates $\widehat{\mu}_1, \widehat{\mu}_2, \ldots$ along with their corresponding quasi-variances $q\widehat{V(\widehat{\mu}_1)}, q\widehat{V(\widehat{\mu}_2)}, \ldots,$ In the second step, these estimates are used to fit the following weighted linear mixed-effect model.

$$\widetilde{y} = X\beta + Zu + \varepsilon \tag{5}$$

Where $\widetilde{y}$ is a vector of genotypic utility mean estimates, $\beta$ and $u$ are vectors of fixed and random effects, with design matrices, $X$ and $Z$ respectively, and $\varepsilon$ is the residual vector. The random effects $u$ and ε follow multivariate normal distributions with mean zero and covariance matrices $\sigma_g^2 I$ and $\sigma_\varepsilon^2 I$ respectively. The vector $u$ contains the random effects of the genotypes.

The mixed model equations are:

$$\begin{bmatrix} X'V^{-1}X & X'V^{-1}Z \\ Z'V^{-1}X & Z'V^{-1}Z + \sigma_g^{-2} I \end{bmatrix} \begin{bmatrix} \beta \\ \widehat{u} \end{bmatrix} = \begin{bmatrix} X'V^{-1}\widehat{\mu} \\ Z'V^{-1}\widehat{\mu} \end{bmatrix} \tag{6}$$

Where $V$ is a diagonal weight matrix whose elements are the inverses of the quasi-variances estimates (qV). In this approach, qV are used instead of variance (V) estimates because they are not tied to a specific label (genotype) and quantify the uncertainty associated with each genotypic mean estimate, rather than the contrasts between pairs of genotypes, as explained in Section 2.2.1. Consequently, defining the matrix $V$ based on qV estimates provides a coherent basis for obtaining valid estimates of the genetic parameters derived from model (5). Comparisons of genetic standard deviation estimates obtained using qV or V as weights for fitting model (3) are presented in Fig. S2.

From the second step onward, the conventional analysis for quantitative traits can be implemented using model (3), enabling the estimation of Best Linear Unbiased Predictions (BLUPs) and variance components. We did not include environmental and year effects in our method, as addressing these factors would have considerably expanded the scope and complexity of this study. However, our approach can be extended to a multi-trial analysis by aggregating tables of genotypic utility mean estimates and their variances in a manner similar to that of Piepho et al. (2012) and Verbyla (2023), thereby enabling the incorporation of environment and year effects.

*2.4.1 Estimation of genetic standard deviation, heritability, and genetic correlations using ranking data*

The genetic standard deviation $\sigma_g$ is estimated using the restricted maximum likelihood (REML). A detailed description of its estimation within the two-stage approach framework can found in Verbyla (2023). Heritability is estimated following Cullis et al. (2006), a method recommended for unbalanced designs, because it effectively accounts for non-constant uncertainty in the estimated genotypic means. Although tricot trials are often conducted using balanced designs,

we anticipated that the variances or quasi-variances of the estimated genotypic means derived from ranking data analysis would not be constant (Frick, 2023). Therefore, within our framework, the approach proposed by Cullis et al. (2006) appears more suitable than other methods for calculating heritability.

The genetic correlation between two environments, and/or two ranked traits, or between a ranked and a quantitative trait can be estimated by extending model (5), as would be done for quantitative traits. The expanded model is presented in Supplementary 1.4.

## 2.5 Simulation study

We evaluate the performance of our framework in estimating genotypic means, their standard errors, genetic standard deviations, and heritability using simulated tricot data. For each plot, yield (in tons per hectare) and corresponding rank-based evaluations are generated. The simulation, estimation, and analysis procedures are described below.

1. We generate observations of maize grain yield in tons/ha following the function,

$$Yield_{jk} = grand\ mean + genotype_{j} + farm_{k} + error_{jk} \tag{7}$$

   Here, $Yield_{ij}$ is the simulated yield of the $j$-th genotype ($j = 1, 2, \ldots, v$) on the $k$-th farm ($k = 1, 2, \ldots, m$) and the $jk$-th plot; *grand mean* is the grand mean of yield; and $genotype_{j}$, $farm_{k}$, and $error_{jk}$ are random deviations for the $j$-th genotype, $k$-th farm, and $jk$-th plot. These deviations are normally distributed with a mean of zero and standard deviations $\sigma_{geno}$, $\sigma_{farm}$, and $\sigma_{res}$, respectively. We define $mean_{j} = grand\ mean + genotype_{j}$ as the true genotypic mean for the $j$-th genotype, and $\sigma_{geno}$ as the true genetic standard deviation of the trial.

2. Within each farm, the three simulated yield values for each plot are ranked from 1 to 3, where 1 is the highest yield and 3 is the lowest yield. Notice that once the simulated observed yields are converted to ranks, the main farm effect is inherently removed from the ranking analysis.

3. We split the data analysis for quantitative metric (yield in tons/ha) and ranking data to estimate genotypic means and their corresponding covariance matrix (first step),

   a. For quantitative metric data on yield, we estimate Best Linear Unbiased Estimates (BLUEs) using the conventional Linear Mixed Model (LMM) approach, treating genotype as a fixed effect, and farm and residual variation as random effects.

   b. For ranking yield data, we estimate the genotypic utility means using the Thurstonian/Plackett-Luce model.

For both analyses, the estimated covariance matrix of the estimated genotypic means was obtained from the observed Fisher information matrix.

In the second step, for both analyses, the genetic standard deviation and heritability are estimated using an LMM, where genotypic means are modeled as a function of a fixed intercept and random effects for genotype and residual variation (see model (5)).

To ensure consistency between analyses based on quantitative and ranking data, the weight matrix used in the second step is always diagonal, with entries equal to the inverses of the quasi-Variances (qV) of the genotypic utility mean estimates. These qV are computed from the observed Fisher information matrix using the R function qvcalc.

To distinguish the estimates obtained from each analysis, we use the term ‘quantitative-based estimates’ for the analysis of the quantitative metric of yield (tons/ha) and ‘utility-based estimates’ for those derived from the analysis of ranking data.

Finally, the quantitative- and utility-based estimates are compared with their reference values. These reference values include the true genotypic means and genetic standard deviation, taken directly from the parameters used in the simulation, while for heritability, these values are obtained from a one-step LMM analysis of yield in tons/ha (Cullis et al., 2006; Smith et al., 2001).

To ensure comparability with utility-based estimates, all parameters and estimates in tons/ha were converted to units of true residual standard deviation ($\sigma_{res}$). Each genotypic mean in tons/ha, either true and estimated, was centered by

subtracting the reference genotypic mean and scaled by dividing by $\sigma_{res}$. Similarly, the genetic standard deviation is divided by $\sigma_{res}$.

We assess the accuracy of our utility-based estimates by comparing them to their reference values using performance metrics and scatterplots. For the estimated genotypic means, we use the Kendall correlation coefficient, which is robust to outliers and sensitive to ordering. Further, we look at bias and Root Mean Square Error (RMSE). For the estimated genetic standard deviation and heritability, the assessment is conducted using scatterplots.

*2.5.1 Simulation of multiple tricot trials using different values of variance components and numbers of replicates*

To test the robustness of our framework, we simulated multiple scenarios of tricot trials with different variance component values and four sample sizes (see Table 1). The values for *grand mean* and $\sigma_{farm}$ were based on previous publications on on-farm yield trials for maize in Africa (Alam et al., 2022; Cairns et al., 2013; Das et al., 2019; Worku et al., 2020). The values for $\sigma_{geno}$ and $\sigma_{res}$ were also derived from the same publications, but their ranges were extended to include both lower and higher extremes values, allowing us to explore the conditions under which reliable estimates can be obtained. The average values for $\sigma_{geno}$ and $\sigma_{res}$ based on these publications are highlighted in bold. The number of genotypes tested was fixed at 20. We simulated 18,480 trials, corresponding to all possible combinations of the parameters in Table 1, (11 values for $\sigma_{geno}$ x; 14 values for $\sigma_{res}$ x; 4 values $n_r$ x 30 simulation replicates). Each trial was simulated and analyzed individually.

*2.5.2 Validation of variances of genotypic utility mean estimates*

We validated the variance obtained from the observed Fisher information matrix (see section 2.2.1), for a simulated trial with parameters $\left(\sigma_{geno} = 0.8, \sigma_{res} = 0.9, n_r = 30\right)$. These parameters are averages of literature-based values, with 30 replicates per genotype, which is common in tricot trials. Using these parameters, we simulated 100 trials with fixed true genotypic means, converting yield evaluations to ranks, and obtaining genotypic utility mean estimates and their variances for each trial. This validation was conducted using the square root of the variances (standard errors, SE). Across simulations, we calculated the standard deviation of the 20 estimated genotypic means and the mean of their SEs, then compared these results in a scatterplot.

## 2.6 Application to observed data

Our observed datasets include tricot trials for groundnut, maize and sweetpotato crops across African regions (Table 2) with both measurements, ranking and quantitative data of yield. In all four datasets, farmers collected the ranking data, while the quantitative data were collected by technicians or trained lead farmers in the sweetpotato and groundnut trials, and by the farmers themselves in both maize trials. The datasets for groundnut and sweetpotato are available on Zenodo (de Sousa, Laporte, et al., 2025a).

In this analysis, the comparison between quantitative- and utility-based estimates followed the same approach as for the simulated data, with the addition of assessing the genetic correlation between the two evaluation methods (as described in Supplementary 1.4) and estimating a heritability derived from rankings calculated directly from the observed yield (tons/ha).

**Table 1. Parameters used in the simulation of tricot trials for maize grain yield (tons/ha).**

| Parameter | Value(s) |
|---|---|
| Grand mean | 3.63 |
| Number of genotypes $(v)$ | 20 |
| Farm standard deviation $(\sigma_{farm})$ | 3.352 |
| Genetic standard deviation $(\sigma_{geno})$ | 0.05,0.1,0.2, 0.4, 0.6, **0.8**, 1.0, 1.2,1.5,1.8,2 |
| Residual standard deviation $(\sigma_{res})$ | 0.4,0.6, 0.8, **0.9**, 1.1, 1.4,1.6,2,2.5,3,3.5,4,4.5,5 |
| Number of replicates per genotype $(n_r)$ | 3, 10, 30, 100 |
| Simulation replicates ($R_{exp}$) | 30 |

Note: 'grand mean' represents the grain mean of yield, $v$ is the number of genotypes tested; $\sigma_{geno}$ is the genetic standard deviation; $\sigma_{farm}$ the location/farm standard deviation, and $\sigma_{res}$ the residual standard deviation. $n_r$ indicates the number of replicates per genotype, and $R_{exp}$ refers to the number of simulation replicates. The averages of literature-based values for $\sigma_{gen}$ and $\sigma_{res}$ derived from publications on on-farm trials are shown in bold.

**Table 2. Description of observed datasets of yield with quantitative and ranking data.**

| Crop | Country | Genotypes | Number of replicates (mean ± CI) |
|---|---|---|---|
| Groundnut | Tanzania | 54 | 28.5±3.46 |
| Maize (Early-maturity) | Kenya | 12 | 66.8±2.70 |
| Maize (Intermediate-maturity) | Kenya | 25 | 58.7±3.95 |
| Sweetpotato | Mozambique | 8 | 16.5±1.69 |

### 2.7 Computational resources and software

We conducted the simulation study and the analysis of observed datasets on a Linux operating system (Ubuntu 20.04.4 LTS) with 251 GiB of memory and 96 cores. Our framework was implemented in the R software (R Core Team, 2023). In the simulation study, we used the *randomize* function from the ClimMobTools package (de Sousa, van Etten, et al., 2025b) to generate a balanced incomplete block design for each trial. For data analysis, we employed the tidyverse package (Wickham et al., 2019), specifically using tidyr for data processing and ggplot2 for data visualization. We used the PlackettLuce (Turner et al., 2020) and gosset (de Sousa et al., 2022) packages to read and process ranking data and used custom-made implementations of Thurstonian and Plackett-Luce models, available in (Dorado-Betancourt & van Heerwaarden, 2025). The optimization of likelihood functions was carried out using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm implemented in the R function optim (Head & Zerner, 1985). BFGS provides an approximation of the Hessian matrix that is required to estimate variances and quasi-variances for the genotypic mean utility estimates. For the implementation of the weighted mixed-effects models, we used the package ASReml (The VSNi Team, 2023).

## 3. Results

The results are presented in three parts: (1) an analysis of estimated genotypic means and standard errors from a simulated tricot trial to illustrate the first-step of the ranking analysis; (2) an assessment of utility-based estimates across multiple simulated tricot trials, with varying genetic and residual standard deviations and numbers of replicates, validating the estimates obtained in both steps of our approach under different conditions; and (3) an application of the proposed method to observed data.

### 3.1 Analysis of estimated genetic values and their standard errors with ranking data

For this analysis, we simulated a trial using the following settings $(\sigma_{geno} = 0.8, \sigma_{res} = 0.9, n_r = 30)$. These values represent the averages of literature-based estimates and the typical number of replicates used in Tricot trials (see Table 1).

Figure 2 shows that the relationship between true and estimated genotypic means is linear across both models used to analyze ranking data. As expected, estimates from the Thurstonian model are unbiased, as the model assumes that rankings reflect underlying utilities that follow a normal distribution, matching our simulation setup. In contrast, the Plackett-Luce model exhibits an expected consistent bias, which is a consequence of its assumption that the underlying utilities linked to the rankings follow a Gumbel distribution, with scale parameter restricted to one. This constraint inflates the variance of the Gumbel distribution when the true utilities are normally distributed, causing the estimated genotypic means to be biased in proportion to their absolute values. The observed bias follows a systematic pattern and can be approximately corrected by applying an appropriate scaling factor. Details are provided in Supplementary 1.2 and 1.3.

Although we explored the identification of a unique scaling factor to align the estimates of both models for any case, we found that this factor is not constant, as it is affected by genetic standard deviation, residual standard deviation, and the number of replicates (see Supplementary 1.3 and Fig. S1). From our simulations, we deduced that when the ratio of genetic to residual standard deviation is 0.71 or lower (as might be expected in on-farm trials, and the number of replicates exceeds 30) the scaling factor typically ranges between 0.85 and 0.90.

Since we are interested in analyzing the performance of the Plackett–Luce model estimates independently of the effect of this scaling factor, all Plackett–Luce estimates were re-scaled to align with the Thurstonian scale (PL utility-based (scl)). The scaling factor was defined separately for each trial using a simple linear regression of Thurstonian estimates on PL estimates, without an intercept.

The estimated standard errors of genotypic means from ranking data reveal three key properties (Figure 3). The first property is that contrary to estimates from quantitative data, utility-based estimates of standard errors are not constant, but

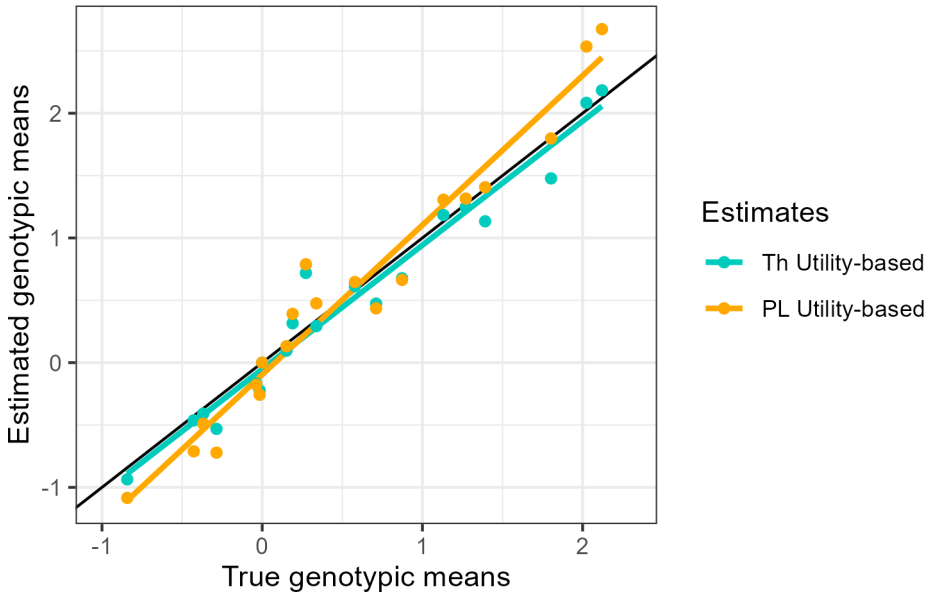



**Figure 2. True versus estimated genotypic means for simulated maize grain yield.** The parameters used in this simulation are $\sigma_{geno} = 0.8, \sigma_{res} = 0.9, n_r = 30$. Both axes are given in standardized units of $\sigma_{res}$. Th and PL refer to estimates obtained using the Thurstonian and Plackett–Luce models, respectively. The solid black line represents the 1:1 line.

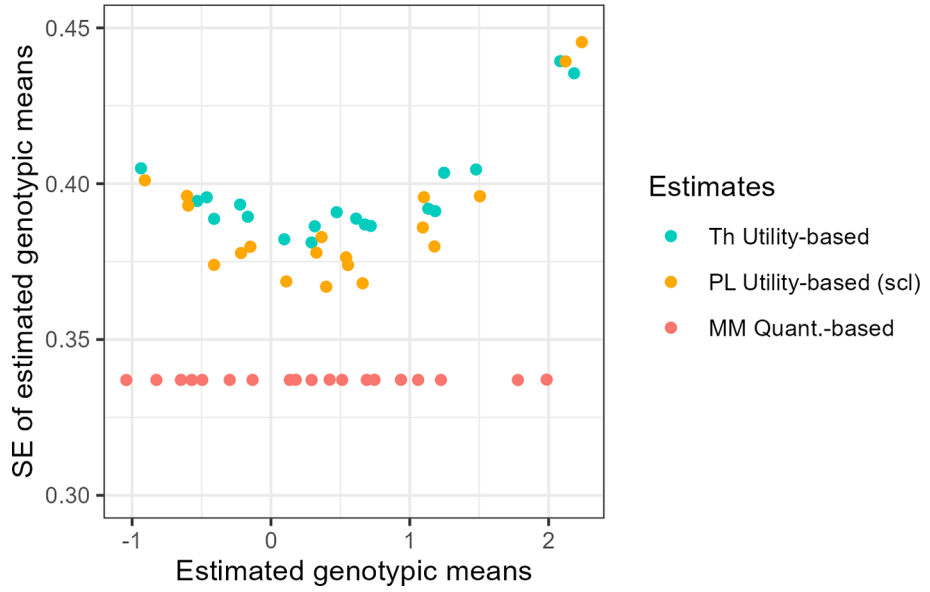



**Figure 3. Estimated genotypic means versus their standard errors (SE) for simulated maize grain yield.** The trial parameters used in this simulation are $\sigma_g = 0.8, \sigma_e = 0.9, r_g = 30$. Both axes are given in standardized units of $\sigma_{res}$. Th, PL, and MM refer to estimates obtained using the Thurstonian, Plackett–Luce, and linear mixed-effect models, respectively. Utility-based estimates from the Plackett–Luce model were scaled to have the same units as the Thurstonian estimates.

follow a quadratic function of the estimated genotypic means. This occurs because when genotypes consistently outperform or underperform others in the sample, less information about their relative values in the latent space can be recovered. The second key property is that standard errors of utility-based estimates are considerably larger than quantitative-based estimates, reflecting the loss of information due to ranking. In fact, we show in Fig. S3 that, for most cases in the expected on-farm trials conditions, about 30% more replicates are needed in ranking trials to achieve the same precision as estimates obtained from quantitative data analysis. The third property is that the scaled standard errors of the

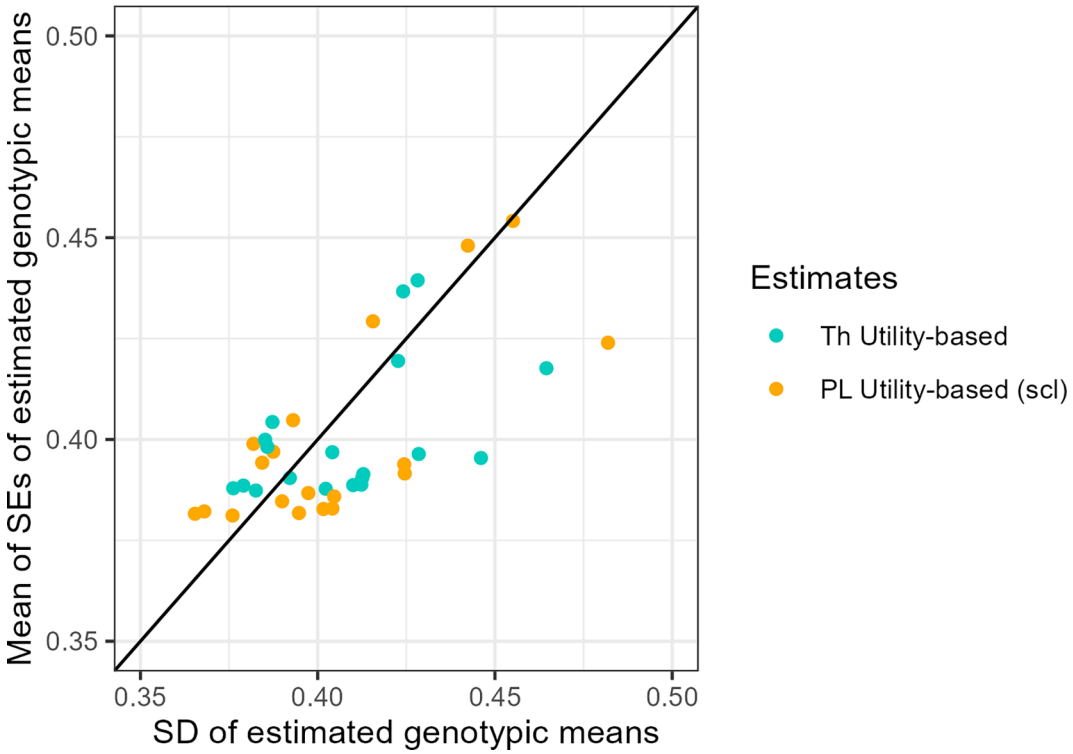



**Figure 4. Expected variability of genotypic means versus observed standard errors for simulated maize grain yield.** Expected variability corresponds to the standard deviation (SD) of estimated genotypic means, while observed variability is the mean of their standard errors (SEs), estimated using the observed Fisher information matrix. The means and standard deviations are calculated over 100 simulation replicates of a simulated trial with parameters $\sigma_{geno} = 0.8, \sigma_{res} = 0.9,\ n_r = 30$, keeping the same true genotypic means. Th and PL refer to estimates obtained using the Thurstonian and Plackett–Luce models, respectively. Utility-based estimates from the Plackett–Luce model were scaled to have the same units as the Thurstonian estimates. The solid black line represents the 1:1 line.

Plackett–Luce model are relatively close to, yet slightly more dispersed than those from the Thurstonian model. These discrepancies may be partly attributable to the scaling factor applied to align the estimates of both models. However, differences in the intrinsic stability of the parameter estimates between models may also contribute.

The validation of standard errors calculated using the observed Fisher information matrix is shown in Figure 4. In this graph, two trials were discarded because they contained undefined estimates with extremely high or low values that affected the calculation of the expected standard error. Cases with undefined estimates are discussed in the following subsection. After removing these two cases, the validation results suggest that the variations inherent in the genotypic utility mean estimates were reasonably well captured. Nevertheless, for some genotypes, particularly those with the highest standard deviations on the right side of Figure 4, the estimation of the standard errors was less precise and probably more replicates are needed.

## 3.2 Performance of genotypic means, genetic standard deviation, and heritability estimates across multiple tricot trials with varying values of variance components and sample size

### 3.2.1 *Genotypic means*

The assessment of the estimated genotypic means in Figure 5 reveals four key insights: (1) After correcting the Plackett-Luce estimates by the scaling factor to align them to the Thurstonian estimates (see section 3.1), the estimates from all three models are unbiased; (2) Utility-based estimates underperform compared to the quantitative-based estimates in terms of precision and correlation. However, this difference in performance is reduced when the number of replicates $(n_r)$ is 30 or more; (3) An increase in genetic standard deviation with low $(n_r)$ leads to higher correlation between true and estimated genotypic means, but decreases the precision of utility-based estimates; and (4) The estimates of the Thurstonian model and the Plackett-Luce (scl) model yield almost the same performance.

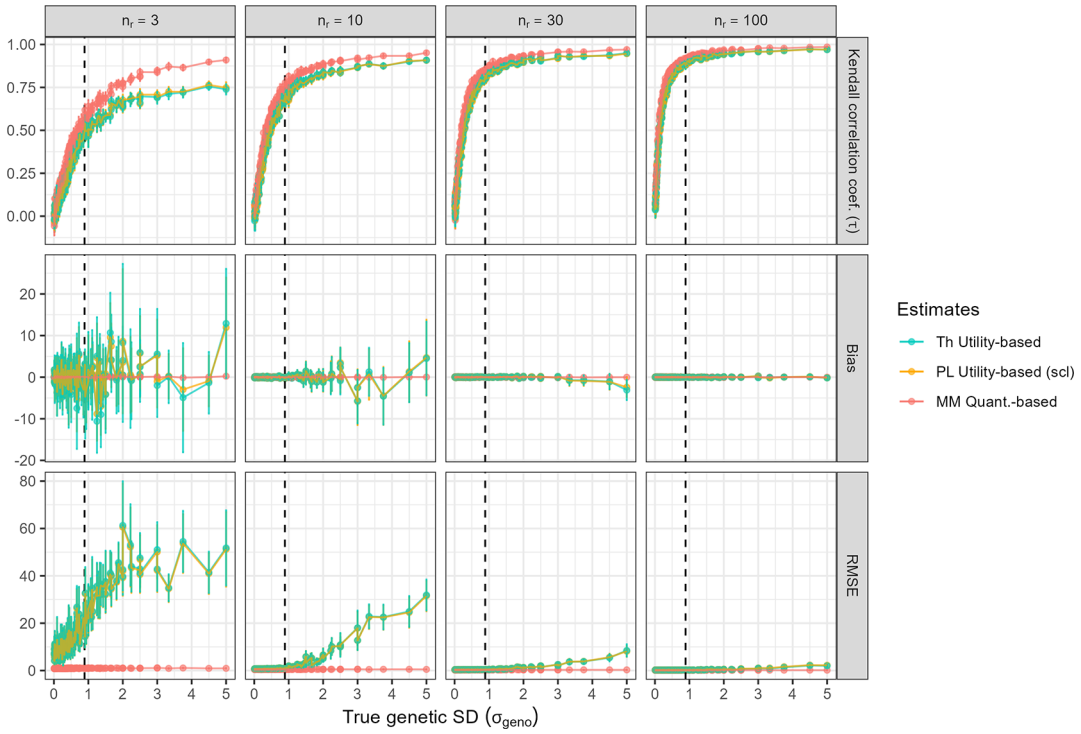



**Figure 5. Genetic variability and replication effects on estimated genotypic performance in simulated maize grain yield.** The top panels correspond to the four possible values for the number of replicates ($n_r$), and the right panels represent the evaluation metrics, calculated based on true and estimated genotypic means, namely, the Kendall correlation coefficient, Bias, and Root Mean Square Error (RMSE), in that order. The y-axis values represent the means of the evaluation metrics obtained over 30 simulation replicates with the same parameter settings, and the corresponding error bars indicate the standard errors of these estimates. Th, PL, and MM refer to values obtained using the Thurstonian, Plackett–Luce, and linear mixed-effects models, respectively. True genetic standard deviation (SD), bias, and RMSE are given in units of $\sigma_e$. Utility-based estimates from the Plackett–Luce model were scaled to match the units of the Thurstonian estimates. Vertical dashed lines highlight the results of a trial with $\sigma_{geno} = 0.8$ and $\sigma_{res} = 0.9$.

From our simulations, we found that 17–19% of the trials produced undefined estimates for the covariance matrix of the genotypic utility mean estimates from the Thurstonian or Plackett-Luce model analyses. These cases are analyzed in Figure 6, which shows that they mainly correspond to trials with high genetic standard deviation ($\sigma_{geno}$ larger than 1.5) and low number of replicates ($n_r$ lower than 30). Under these conditions, it is more likely that certain genotypes are repeatedly ranked as either consistently best or worst at farms. This invariance in ranking results in a lack of information

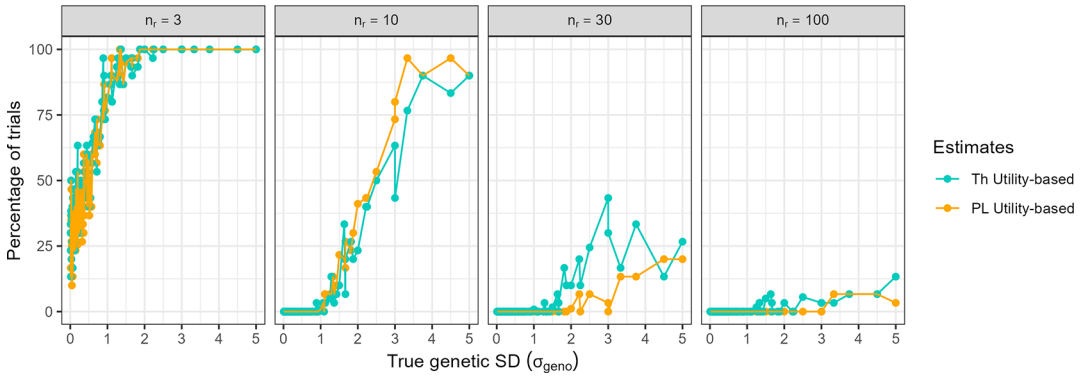



**Figure 6. Genetic variability and replication effects on trials with undefined covariance matrix.** Th and PL refer to estimates obtained using the Thurstonian and Plackett–Luce models, respectively. True genetic standard deviation (SD) is reported in units of $\sigma_{res}$.

needed to estimate continuous relative differences among genotypes. In the most extreme case, a genotype that is always ranked either first or last provides no variability in its relative continuous distance in the latent space, making it impossible to accurately estimate its genotypic value. As a result, the estimate may become extremely large (positive or negative), and the associated variance may also be very large or infinite. A detailed example is provided in Fig. S4.

### 3.2.2 *Genetic standard deviation and heritability*

In contrast to Figure 5, Figure 7 shows that accurate estimates of genetic standard deviation can be obtained when genotypic means are accurately estimated. The performance of these estimates is negatively affected as the true standard deviation increases, often leading to underestimation. In some cases, it is not possible to estimate values when the number of replicates is too low. As mentioned, these conditions produce undefined genotypic utility means estimates, resulting in undefined or inflated estimates for their respective variances. High variance estimates lead to high quasi-variances, which serve as weights in the mixed model for estimating genetic standard deviation. Consequently, genotypes with undefined estimates contribute little or nothing to the estimation of genetic standard deviation, explaining the observed underestimation. Although this loss of performance can be mitigated by increasing the number of replicates in trials with true genetic standard deviations above 2, more than 100 replicates may be required to obtain reliable estimates.

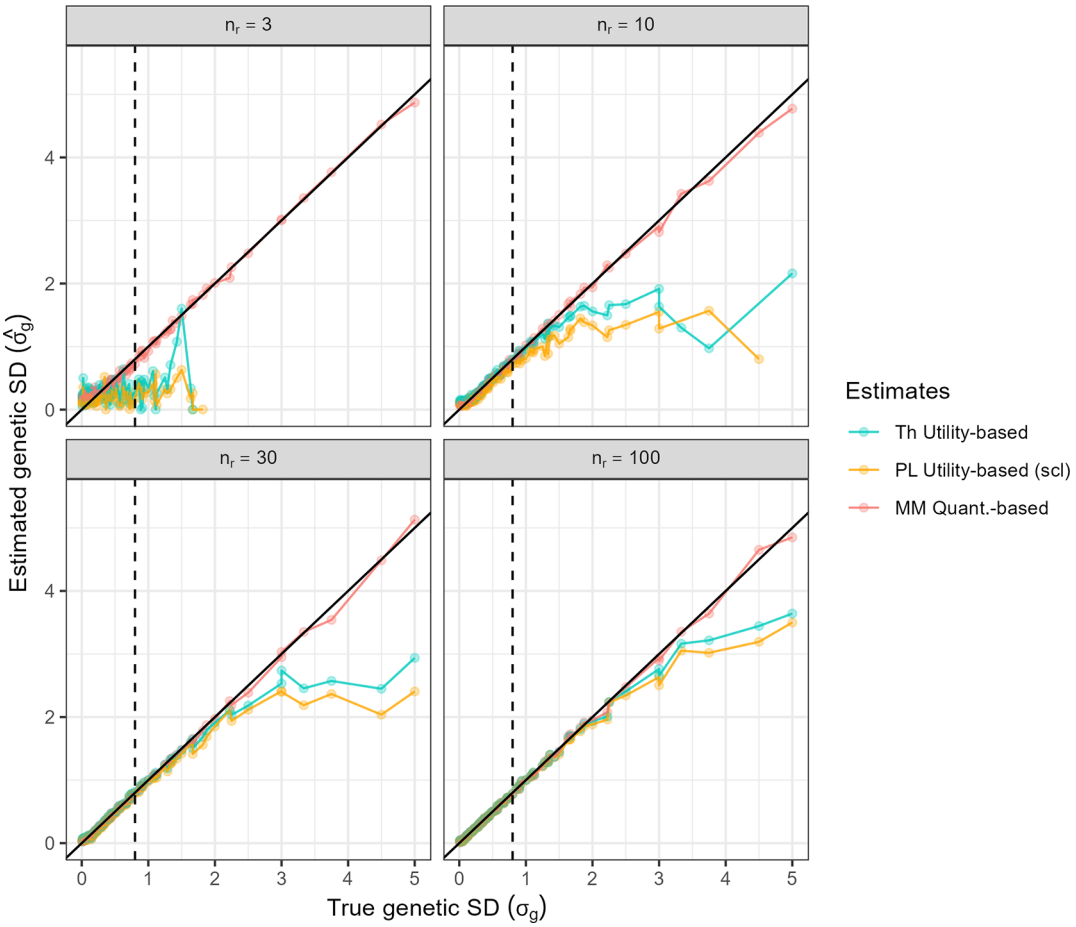



**Figure 7. True versus estimated genetic standard deviation (SD) in simulated maize grain yield trials.** The top panels correspond to the four possible values for the number of replicates ($n_r$). The y-axis values represent the means of the estimates obtained from over 30 simulation replicates with the same parameter settings, (the disaggregated version of this figure is in Fig. S5). Both axes are given in standardized units of $\sigma_e$. Th, PL, and MM refer to estimates obtained using the Thurstonian, Plackett–Luce, and linear mixed-effects models, respectively. Utility-based estimates from the Plackett–Luce model were scaled to match the units of the Thurstonian estimates. The solid black line represents the 1:1 line. Vertical dashed lines highlight the results of a trial with $\sigma_g = 0.8$ and $\sigma_e = 0.9$.

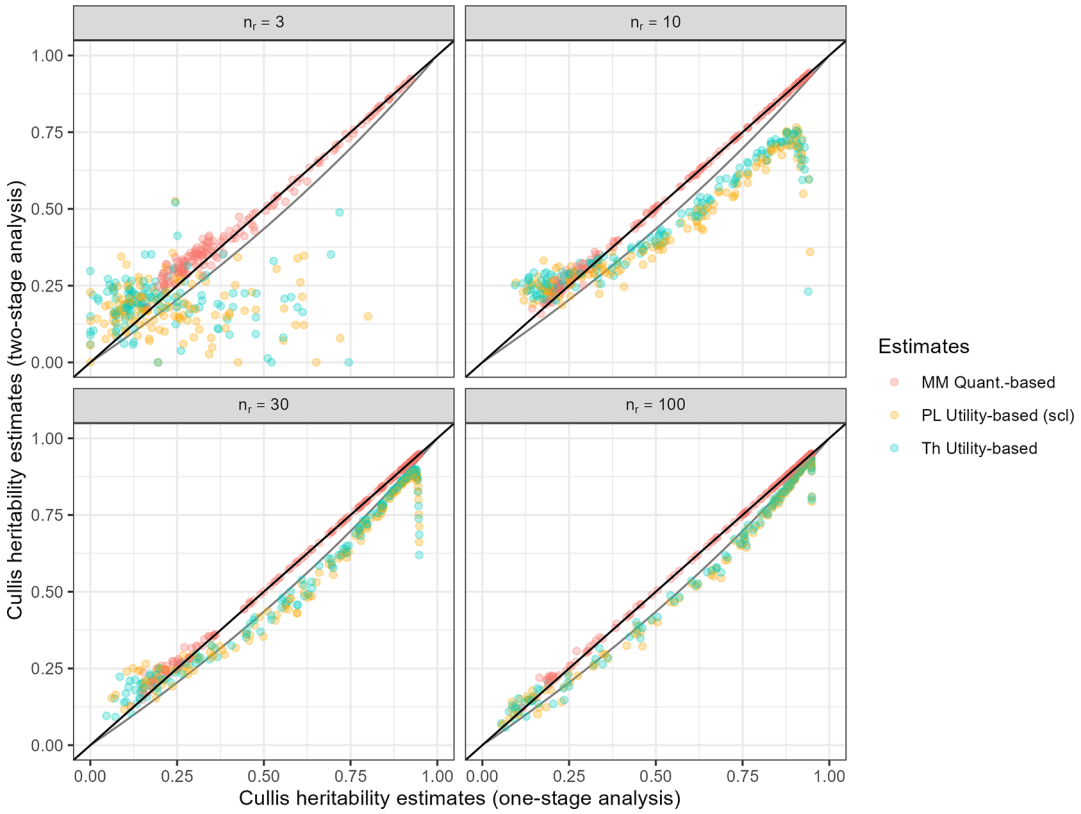



**Figure 8. Reference versus estimated Cullis heritability for simulated maize grain yield.** Reference values are calculated using a one-step analysis on yield (in tons/ha), while estimated values are calculated using a two-step approach. The top panels correspond to the four possible values for the number of replicates ($n_r$). The x- and y-axis values represent the means of the estimates obtained over 30 simulation repicates with the same parameter settings (the disaggregated version of this figure is in Fig. S6; simulated trials with heritability estimates $\leq 0$ were discarded. Th, PL, and MM refer to estimates obtained using the Thurstonian, Plackett–Luce, and linear mixed-effects models, respectively. The dashed black line represents the 1:1 reference line. The solid line is the expected values for utility-based heritability.

The calculation of Cullis heritability yielded negative values in 17–19% of all simulated trials. These negative estimates were present across all mixed-effects models used for quantitative data analysis, as well as in Thurstonian and Plackett-Luce analyses, which suggest that this is a limitation of this method of estimating heritability. Such cases were particularly associated with extremely low genetic standard deviations combined with high residual standard deviations. These cases were excluded from the analysis.

Figure 8 shows an overestimation for reference values of heritability lower than 0.25, affecting both estimates with quantitative data and utility-based estimates. Moreover, we observe that 3 replicates are insufficient to obtain reliable utility-based estimates. When the number of replicates are 10 or more, the relationship between estimated and reference heritability values becomes linear and close to one-to-one for quantitative-based estimates, whereas it follows an exponential pattern for utility-based estimates. The expected utility-based heritability (solid line) is calculated from the entry-mean broad-sense heritability, adjusted to account for the variability in ranking data as $\left(\hat{\sigma}_g^2 / \left(\sigma_g^2 + 1.3 * \sigma_\varepsilon^2 / n_r\right)\right)$, where the factor 1.3 adjusts for the difference in precision between utility- and quantitative- based measurements (Fig. S3). Notably, utility-based estimates of heritability are generally lower than quantitative-based estimates, which is expected given the greater uncertainty associated with rankings. The gap between quantitative- and utility-based estimates decreases as the number of replicates increases, since the high uncertainty inherent to ranking comparisons is compensated for by a larger sample size.

### 3.3 Observed data example

Table 3 shows that genetic correlations in all four cases were high and close to 1, indicating that both quantitative yield evaluations and yield rankings reflect the same underlying genetic effects. For groundnut and early-maturity maize,

**Table 3. Estimates of genetic parameters from quantitative and rank-based data for the observed yield in tons/ha.**

| Crop | $\widehat{r}_{g^{qt},g^{ut}}$ | $\widehat{\sigma}_{g^{qt}(SC)}$ | $\widehat{\sigma}_{g^{ut}}$ | $\widehat{H}^2_{qt}$ | $\widehat{H}^2_{ut}$ | $\widehat{H}^2_{ut_2}$ |
|---|---|---|---|---|---|---|
| Groundnut | 0.961 | 0.179 | 0.161 | 0.359 | 0.229 | 0.218 |
| Maize (Early-maturity) | 0.997 | 0.204 | 0.205 | 0.607 | 0.525 | 0.581 |
| Maize (Intermediate-maturity) | 0.981 | 0.033 | 0.170 | 0.001 | 0.407 | 0.297 |
| Sweetpotato | 0.978 | 0.201 | 0.145 | 0.324 | 0.136 | 0.147 |

*Note* : $\widehat{r}_{g^{qt},g^{ut}}$, denotes the genetic correlation between genotypic effects estimated from quantitative and rank-based data. The parameters $\widehat{\sigma}_{g^{qt}(SC)}$ and $\widehat{H}^2_{qt}$ represent the genetic standard deviation (standardized in units of $\widehat{\sigma}_{res}$) and Cullis heritability, respectively, both estimated from yield measured in tons/ha. $\widehat{\sigma}_{g^{ut}}$ and $\widehat{H}^2_{ut}$ correspond to the genetic standard deviation and Cullis heritability derived from observed rankings. $\widehat{H}^2_{ut_2}$ refers to Cullis heritability estimated from direct rankings based on yield in tons/ha. All estimates obtained from ranking data correspond to utility-based estimates derived from the Thurstonian model.

genetic standard deviation estimates were relatively similar, as expected from our simulation analysis. In contrast, for sweetpotato, differences among genotypes were more evident when using quantitative data than when using ranking data, resulting in a higher estimate for genetic standard deviation of the former. Heritability estimates for groundnut, early-maturity maize, and sweetpotato were consistently higher when derived from quantitative data ($\widehat{H}^2_{qt}$), reflecting the expected loss of information when using rankings ($\widehat{H}^2_{ut}$). Furthermore, heritability estimates based on farmers' rankings ($\widehat{H}^2_{ut}$) were nearly identical to those obtained from rankings derived directly from quantitative yield data ($\widehat{H}^2_{ut_2}$), suggesting that farmers' perceptions or ranking skills did not bias these heritability estimates. The most distinct result was observed for intermediate-maturity maize, where the genetic standard deviation estimated from quantitative data was too low, leading to a very low heritability estimate ($\widehat{H}^2_{qt}$), whereas the ranking-based analysis yielded higher genetic standard deviation and positive heritability estimates.

## 4. Discussion

Ranking comparisons are a practical and reliable way for small-scale farmers to evaluate genotypes on farm. However, analytical methods for this type of data remain limited in the field of plant breeding. In this study, we develop and present a novel procedure that plant breeders can use for quantitative genetic analysis using ranking data, as a base for a more sophisticated multi-trial analysis. By using simulated and observed data, we demonstrated that our method can generate reliable and robust estimates of genotypic means, their variances, genetic standard deviation, and heritability. These estimates remain accurate as long as the number of replicates (generally around 30 replicates) used in the trial compensates for the uncertainty caused by a high ratio of genetic to residual standard deviation. These conditions are reasonable and feasible to achieve in on-farm testing, where this ratio is expected to be low.

With our procedure, ranking evaluations for both quantitative and non-quantitative traits collected on-farm can be processed and analyzed in a similar way to conventional quantitative traits, offering a straightforward, cost-effective, efficient, flexible, and easy-to-implement approach. Moreover, this procedure can be adapted and extended to multi-trial analyses following the two-step approach developed for quantitative trait analyses in plant breeding (Möhring & Piepho, 2009; Smith et al., 2001; Verbyla, 2023). Its combination of adaptability to a quantitative genetic framework and user-friendly implementation in analytical tools could encourage the plant breeding research community to adopt it in their pipelines.

Based on our simulation study, we corroborate that reliable estimates of genotypic means and their standard errors can be obtained from ranking-data analyses. This is feasible when rankings are assumed to be linked to continuous underlying utilities, which is a property inherent to both the Thurstonian and Plackett–Luce models. The accuracy of these estimates depends on the genetic-to-residual standard deviation ratio and on the number of replicates per genotype. In most cases, 30 replicates were sufficient; however, a larger number of replicates was required when this ratio exceeded 2. Assuming a normally distributed underlying trait, only the Thurstonian model yielded accurate estimates of genotypic utility means, whereas the Plackett–Luce model exhibited a systematic bias. Nevertheless, if the goal is only to determine the overall ranking (i.e., the order) of genotypes, both models perform similarly.

The parameter constraints applied to the Thurstonian model allow the utilities to be interpreted in units of the residual standard deviation of the trial. This property enables a direct correspondence between utility- and quantitative-based estimates for the same trait. Such comparability is particularly advantageous when integrating information from trials involving different evaluation types (i.e., rankings versus quantitative measures), as the residual standard deviation can often be derived from standard protocols or historical data. Although similar constraints are applied to the Plackett–Luce model, the resulting genotypic utility means do not scale directly to the residual standard deviation due to the systematic bias noted above. We demonstrated that this bias can be corrected through a scaling factor; however, this factor is not constant and varies according to the genetic-to-residual standard deviation ratio and the number of replicates (Fig. S1). When the ratio is below 1 and approximately 30 replicates per genotype are available, the expected scaling factor for the Plackett–Luce estimates is approximately 0.9. After this adjustment, both models perform similarly in estimating genotypic means and genetic standard deviation.

The assessment of model performance further revealed that, unlike the standard errors obtained from an analysis of quantitative traits under a balanced design, those from both the Thurstonian and Plackett–Luce models are not constant and follow an almost exponential pattern, as previously reported by Bürkner et al. (2019) and Frick (2023). Furthermore, the standard errors derived from ranking data were substantially larger than those obtained from metric data, reflecting the expected loss of information. Under typical on-farm conditions, approximately 30% more replicates would be required to achieve a precision equivalent to that obtained from quantitative measurements. While this may seem discouraging, it is essential to realize that rankings have distinct advantages in data collection from both economic and logistical perspectives since they are easier to collect and do not require specialized instruments.

Our results confirm that the second step of our approach can accurately estimate genetic standard deviations, which is feasible using a weighted linear mixed-effects model. This requires that the model be fitted using accurate genotypic utility mean estimates and their associated variances, which, as mentioned above, is feasible when the experiment includes a sufficient number of replicates. By obtaining precise estimates of genetic standard deviation, we validated that the non-constant heterogeneity observed in the genotypic utility mean estimates (See Figure 3) is properly accounted for by using the inverse of the quasi-variances as weights in the mixed model. In addition, we demonstrate that this model can be used to estimate heritability following the method of (Cullis et al., 2006). The resulting estimates are consistent with our expectations, as those derived from utilities (based on ranking data) were lower than those obtained from quantitative data, consistent with previous findings (Ricard & Legarra, 2010). We interpret this as evidence that the use of evaluations based on ranking data entails a loss of information, leading to lower heritability estimates. Finally, the fact that reasonable heritability estimates from utilities were obtained using the method of (Cullis et al., 2006) validates that the prediction error variances were adequately represented in our mixed model.

The analysis of observed data demonstrated consistency and agreement between yield evaluations using rankings provided by farmers and quantitative measurements taken by technicians, and in some cases for farmers. This was reflected in high genetic correlations, indicating that genetic performance and differences among genotypes were similarly expressed across both evaluation methods. Sweetpotato had the lowest number of replicates, and yield data were measured by technicians. Under these conditions, it is expected that differences among genotypes would be more clearly reflected in the estimation of genetic standard deviation when using quantitative data than when using ranking data. Interestingly, the weights of early-maturity maize were taken by farmers, but the high number of replicates likely increased the precision of the estimated genetic standard deviation using ranking data, making it comparable to the one obtained from quantitative data (farmer weights). The trial for intermediate-maturity maize showed the lowest quantitative estimates for genetic standard deviation and heritability. This could be explained by the greater exposure of these genotypes to late-season stresses, such as early cessation of rainfall or heat stress, compared with early-maturity genotypes. Because genetic variation was very low, we observed that estimates based on ranking data were overestimated, highlighting the limitation of cullis heritability under these particular conditions, which will be discussed in the caveats paragraph below. To conclude, results from observed data demonstrate that quantitative genetic analysis based on ranking data is feasible. The rankings provided by farmers are reliable and consistent with quantitative measurements taken by technicians, effectively reflecting farmers' preferences and needs. Thus, our procedure offers breeders a useful tool to analyze this type of information.

It is known that small-scale farmers can accurately measure quantitative traits, which makes a ranking approach complementary or, in some cases, unnecessary for evaluating genotypes (Nabateregga et al., 2025; Paliwal & Jain, 2020). In practice, our procedure could be particularly useful in experiments that involve other stakeholders, such as consumers or traders, and focus on the evaluation of non-quantitative traits, including preferences for color, flavor, overall preference or marketability. These traits are key for the adoption of new genotypes, especially for small-scale holders.

A caveat identified in our approach arises when certain genotypes are consistently ranked at either extreme (highest or lowest) across all observations. This situation leads to undefined estimates of genotypic utility means and inflated variances, thereby reducing the accuracy of genetic standard deviation and heritability estimates. Such cases mainly occur in trials with a high true genetic-to-residual standard-deviation ratio and a limited number of replicates. Because our approach is intended for on-farm testing, where genetic variability is expected to be less expressed under untested field conditions and experimental variability is higher, we anticipate this ratio to remain below 2. Under these circumstances and based on our simulation study, approximately 30 replicates per genotype are sufficient for reliable estimations. We do not recommend removing these "problematic" genotypes, as doing so could underestimate genetic standard deviation. A more appropriate solution is to increase the number of replicates per genotype or restrict the analysis to the ranking (order) of the genotypic utility means estimates. As demonstrated by the Kendall correlation coefficients (see Figure 5), the ranking of itself is not affected by this limitation.

We identified two limitations related to the study that should be considered: (1) Only yield was analyzed in both the simulated and observed datasets, as it is typically assumed to be continuous and normally distributed. In fact, most traits with commercial value exhibit continuous distributions and result from the joint segregation of multiple genes, which generally leads to an approximately normal distribution (Falconer, 1996). We acknowledge that some traits may have unknown probabilistic distributions. In these cases, it is worth evaluating whether the Central Limit Theorem ensures normality of the estimated genotypic utility means; and (2) although we consider the Cullis et al., (2006) method to be the most effective approach for estimating heritability in the analysis of trials with ranking data, we are aware that it can be imprecise and may yield unstable or negative estimates when the ratio of genetic to residual standard deviation is extremely low, such as the case of intermediate-maturity maize. Other methods for estimating heritability could be explored in the future. In Fig. S7, we calculate broad-sense heritability using the median of the variances of the genotypic utility mean estimates.

This study provides a basis for the analysis of ranking data in plant breeding. We suggest that future work may focus on: (1) extending our method to include environmental and year effects and their interaction with genotypic effects; (2) estimating and validating additional parameters such as genetic gain and genetic correlation between environments and/or traits; (3) developing and testing a method for utility-based multi-trait analysis that enables the simultaneous evaluation of varying traits; (4) improving utility-based estimates by incorporating data of farms where evaluations using both rankings and quantitative data have been conducted; and (5) assessing the performance of the estimations when the data include comparisons involving more than three genotypes and/or allow for ties.

Together, our results and those reported in the literature illustrate the potential of ranking data in breeding programs. In particular, we showed a method with potential of generating reliable on-farm estimates that integrate farmers' preferences by analyzing tricot data.

## Statements and declarations

### Ethics approval

The study involved voluntary participation of farmers who provided feedback on the tested genotypes. According to institutional and national guidelines for non-interventional social research, formal ethical approval was not required.

### Consent to participate

All participants were informed about the study objectives and provided verbal/written informed consent.

## Availability of data and code

The observed data for groundnut and sweetpotato used in this study are publicly available and can be accessed at: Global multi-crop agricultural trial data supported by citizen science, Zenodo https://doi.org/10.5281/zenodo.17112492, and are released under the Creative Commons Attribution–ShareAlike 4.0 International License (CC BY-SA 4.0). The observed maize data, however, were subject to CIMMYT data governance procedures and, at the time of publication, were still being prepared for open access; these data were therefore available from the authors upon reasonable request.

The R functions and simulation workflow used in this study are publicly available at:

- Source code available from: [https://github.com/hdorado/tricot-ranking-analysis]
- Archived software available from: [https://doi.org/10.5281/zenodo.17942919]
- License: [MIT License]

Supplementary Information includes two sections (1. Supplementary Material and Methods and 2. Supplementary Figures), which are available via the Zenodo repository https://doi.org/10.5281/zenodo.17941054 and under the terms of the Creative Commons License (CC BY 4.0),

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